

Motivations and Introduction

DATABASES — Lezione 1

Data is important

Information age

- We live in the information age.
- Every minute of the day, a large amount of data is generated online.
- According to an estimation, a typical individual creates between 50,000 to 100,000 data points daily.
- Data constitutes the basic infrastructure of our lives, making things simpler, possible, better (and worse).

Examples

- Knowing when is the next bus is.
- Knowing in which room the DB lesson is.
- Knowing whether an invoice was paid.
- Being able to retrieve the results of my latest health check.
- Knowing ...

Why data matters

Informed Decision-Making	Data allows to see what actually works; data allows to see the effects of policies and actions.
Personalization and Convenience	Algorithms use data to tailor experiences (for instance: movie recommendations, fastest route to avoid traffic).
Scientific Discovery, Problem Solving, Healthcare	Some disciplines rely heavily on data (track diseases, develop new medicines, model climate change).
Efficiency and Automation	Data helps streamline operations, optimize supply chains, and automate repetitive tasks, saving time and resources across industries.

Example: Managing a Company

Areas to take care of to manage a company:

- Client and Prospects
- Orders and Sales
- Inventory, Production,
- Quotations, Invoicing, Accounting
- Projects and performance
- Human Resources, Presence, Timesheets
- Knowledge & Documents

Approaches:

- On people's computers
- On shared folders
- On a shared system

Data is important: Example

- Profit & Loss projection
- Systems performance tracking
- Projection of inventory lock stock alerts
- User engagement and experience, Retention metrics
- Which products actually work, which functions actually work

Data is important to make informed decisions

- 2006, Clive Humby: «Data is the new oil»
 - Raw data has little value by itself
 - Refining unlocks insights
- From data to structured information
 - We need a «refinery»

Databases

Database A structured collection of data stored in a computer system, managed by specialized software (DBMS) for efficient, reliable, and multi-user access.

DBMS: Database Management System

Managing large and complex data is difficult

In the early days, applications to store data were built directly on top of file systems.

File-system problems

Data redundancy and inconsistency	Data is stored in multiple file formats resulting in duplication of information in different files.
Difficulty in accessing data	Need to write a new program to carry out each new task.
Data isolation	Multiple files and formats.
Integrity problems	Integrity constraints (e.g., account balance > 0) become «buried» in program code rather than being stated explicitly; hard to add new constraints or change existing ones.

Atomicity, concurrency, security

Atomicity of updates	Failures may leave database in an inconsistent state with partial updates carried out. Example: transfer of funds from one account to another should either complete or not happen at all.
Concurrent access by multiple users	Concurrent access needed for performance; uncontrolled concurrent accesses can lead to inconsistencies. Ex: two people reading a balance (say 100) and updating it by withdrawing money (say 50 each) at the same time.
Security problems	Hard to provide user access to some, but not all, data.

How big? How Complex?

Walmart's Retail Link database manages its supply chain and inventory management:

- Inventory levels
- Sales data
- Supply chain logistics

Used to:

- Assess product performance
- Inventory replenishment
- Pricing strategies
- Demand forecasting

Scale:

- Billions of transactions per day
- Integrating data from thousands of suppliers and millions of products

Fun story about Walmart: Strawberry Pop-Tarts increase in sales, like seven times their normal sales rate, ahead of a hurricane. And the pre-hurricane top-selling item was beer.

How we use databases at Shair.Tech

- As the basic infrastructure of our products (BringTheFood)
- As the data storage for monitoring applications (Analytics and Statistics)
- As the data management systems for business information (Redmine, CRM)
- As an efficient system to store and extract data for reporting purposes

Databases at Shair.Tech

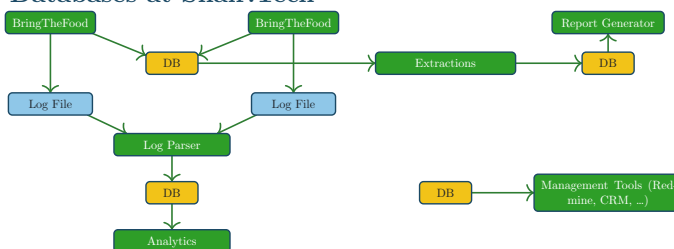


Fig. 1: Data-flow architecture: system components and their databases.